Bohdan Bozhenko

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personal website | github | linkedin
Right to work in the UK - no sponsorship required.

SUMMARY

Recent BSc (Hons) Computing graduate (First-Class Honours) from Glasgow Caledonian University with hands-on experience building full-stack applications in React/Next.js, FastAPI and Spring Boot, including real-time and AI-integrated systems running on local LLMs. Strong focus on backend engineering, testing and secure design. Based in Dumfries, open to relocation and available for immediate start.

WORK EXPERIENCE

Volunteer - The Devil's Porridge Museum
Dumfries, UK | Dec 2024 – June 2025

- Maintained and enhanced the museum's WordPress website, restructuring pages and adding content to improve usability and visitor engagement
- Drove event enquiries to more than double the initial target (50+ additional participants) through improved online visibility and targeted digital outreach
- Supported event operations by coordinating schedules, assisting promotions and managing digital communication across online channels

PROJECTS

AI Office - **[GitHub]**

Next.js · React Three Fiber · FastAPI · PostgreSQL · Ollama · Docker

- Built a local multi-agent app where AI agents on a 27B local model work through staged tasks (plan, web research, draft, self-check) with a human review queue, shown live in a 3D office
- Designed a GPU request queue, crash recovery and guarded web tools (address checks after DNS resolution, re-checked redirects) to keep agents reliable and safe
- Audited it for security and load with two Claude Code agents (33 findings); fixed 14 with regression tests, fully closing 4 of 6 High and mitigating the other 2; 483 backend tests in GitHub Actions CI

SearchWork - **[GitHub]** | **[Demo]**

Next.js · FastAPI · PostgreSQL · JWT · Google OAuth

- Built and deployed a job-tracking platform that aggregates listings from 11 job-board APIs (including Adzuna and Reed), with Claude-powered CV parsing and match scoring; used daily for my own job search
- Implemented JWT authentication with refresh tokens and Google OAuth, plus rate limiting and CORS hardening
- Ran a security audit and fixed 12 findings, including hardcoded credentials and missing authorisation checks; tested mobile flows with Playwright

Programming Helper AI - **[GitHub]** | **[Demo]**

Next.js · React · TypeScript · Tailwind · OpenAI API

- Built an AI-powered learning platform providing real-time code explanations and contextual suggestions
- Designed interactive problem-solving workflows to improve user engagement and learning efficiency
- Integrated OpenAI API for dynamic response generation and structured prompt handling

Real-time Chat Application - [GitHub]

Spring Boot · WebSocket (STOMP) · React · PostgreSQL

- Built a group chat over STOMP/WebSocket with a SockJS fallback and message history in PostgreSQL
- Used a persist-then-broadcast flow and per-session ordered frame handling, so every broadcast message is already saved
- Rebuilt the backend on Spring Boot 3.4, replacing the earlier Node.js/Express version; 23 tests, including Testcontainers integration tests, run in GitHub Actions CI

FLYON - [GitHub]

Next.js · Node.js/Express · PostgreSQL/PostGIS · WebSocket · TypeScript

- Built a flight analytics platform for drone and FPV pilots with real-time telemetry over WebSocket, a flight health score and danger-zone warnings
- Integrated Betaflight Blackbox logs and added KML/GPX export; drone-agnostic design driven by telemetry input
- Traced a slow telemetry pipeline to quadratic processing despite full indexing and fixed it with an incremental accumulator: 9,000-point ingest dropped from 152.8 s to 50.1 s, with per-point latency flat at ~4 ms; wrote it up for HackerNoon (featured on the homepage)

EDUCATION

BSc (Hons) Computing - Glasgow Caledonian University
Glasgow, UK | Sep 2023 - Jun 2026
First-Class Honours (1st)

Relevant modules: Software Engineering & OOP, Data Structures & Algorithms, Web Application Development (I & II), Computer Networks & Cloud Computing, DevOps, Secure Software Development, Honours Project (Programming Helper AI)

HNC Cyber Security (SCQF Level7) - Dumfries and Galloway College
Dumfries, UK | Aug 2022 - Jun 2023

Focused on cybersecurity fundamentals, system vulnerabilities, secure networking, and data analysis.

Junior Bachelor's Degree in Software Engineering - National Aerospace University "KhAI"
Kharkiv, Ukraine | Sep 2021 - Jun 2023

Covered data structures & algorithms, .NET development, SQL databases, software testing, and security fundamentals.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Java, SQL
Frontend: React, Next.js, React Three Fiber, HTML, CSS, Tailwind
Testing: pytest, Playwright, Vitest
Backend: FastAPI, Spring Boot, Node.js, REST APIs, WebSocket/STOMP, SQLAlchemy, Alembic
Databases: PostgreSQL, MySQL, MongoDB
DevOps & Cloud: Docker, CI/CD, Jenkins, AWS
Tools: Git, GitHub, Postman, IntelliJ IDEA, VS Code, PyCharm
AI/LLM APIs: Claude API (Anthropic), OpenAI API, Ollama, Faster Whisper
AI Dev Tools: Claude, Claude Code, GitHub Copilot, Cursor

PUBLICATIONS

- "I Added Every Index, but My Telemetry Pipeline Was Still Quadratic" - HackerNoon, 2026 (featured on the HackerNoon homepage)
- "I Built an AI Job Tracker While Job Hunting: The AI Was the Easy Part" - HackerNoon, 2026

CERTIFICATES

- Building with the Claude API - Anthropic, 2026
- Cisco Networking Essentials - Cisco, 2023
- The Complete JavaScript Course 2025 - Udemy, 2026

LANGUAGES

English - Professional working proficiency
Ukrainian - Native